

D019

Data Literacy and Evidence-Based Practices

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A school improvement team meeting in the newly renovated media center reviews results from a spring survey asking families how welcome they feel at school. The survey was offered in three languages. What type of data is this?

- A. Student learning data.
- B. Perception data.
- C. Demographic data.
- D. School process data.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A fourth-grade teacher who recently finished a master's degree wants to decide which students need reteaching on fractions tomorrow. She has last spring's state test results, today's exit tickets, and the district's demographic report, which was just updated. Which data source fits the decision best?

- A. Today's exit tickets.
- B. All three equally, since more data is always better.
- C. The updated demographic report.
- D. Last spring's state test results.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A seventh-grade teacher at a school with a new bell schedule reports quiz scores of 72, 75, 78, 80, 81, 83 and 20. The 20 is from a student who was absent most of the unit. Which number best describes a typical score for this class?

- A. 78, the median.
- B. 20, since it is the lowest score.
- C. 63, the range.
- D. About 69.9, the mean.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Using the same seven scores (72, 75, 78, 80, 81, 83, 20), the teacher removes the 20. Which statistic changes more, and by about how much?

- A.** The mean, rising by 20 points.
- B.** The mean, rising by about 8 points (69.9 to 78.2).
- C.** The median, rising by about 8 points.
- D.** Neither changes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

School A, which has a new STEM lab, had 80% of students proficient last year and 80% this year. School B went from 40% to 55%. Both enroll about 600 students. Which school shows more evidence of student growth?

- A.** School A, because it has a STEM lab.
- B.** School A, because its proficiency is higher.
- C.** Neither, since they have the same enrollment.
- D.** School B, which rose 15 percentage points.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A board member says, “Our highest-scoring school must have the best teaching, since 92% of its students are proficient.” What does the data leave out?

- A.** The percent not proficient, 8%.
- B.** Nothing; high proficiency proves strong teaching.
- C.** How much students grew; they may have entered already proficient.
- D.** The school’s enrollment size.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Students who chose to attend a voluntary after-school tutoring program, held in the recently expanded library, scored 12 points higher on the spring test than those who did not attend. The principal wants to credit the program. The tutors were all retired teachers. What is the main caution?

- A. A 12-point gap is too small to matter.
- B. The retired tutors make the results unreliable.
- C. The program clearly caused the 12-point gain.
- D. Students who volunteer may differ in motivation or support, so the gap may not be caused by the program.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A principal wants stronger evidence that a new reading intervention causes improvement. Which design helps most?

- A. Survey teachers on whether they liked it.
- B. Randomly assign similar students to receive the intervention or not, and compare gains.
- C. Report the average score of participants only.
- D. Compare volunteers with non-volunteers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A parent who teaches college statistics sees her son’s report showing “60th percentile” on a nationally normed reading test taken on the school’s new laptops. She asks whether that means he got 60% of the questions right. What is the accurate explanation?

- A. He met 60% of the grade-level standards.
- B. The score is unreliable because the test was on laptops.
- C. He scored higher than about 60% of students in the national comparison group.
- D. He answered 60% of the questions correctly.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A state test reports a student as “Meets Standard” because her score passed the fixed cut score. What kind of interpretation is this, and what does it NOT tell you?

- A.** Criterion-referenced; it does not tell how she ranks compared with other students.
- B.** Criterion-referenced; it tells her exact percentile.
- C.** It tells everything about her reading ability.
- D.** Norm-referenced; it does not tell whether she met the standard.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Ninth-grade failure rates rose this year at a high school with a new football stadium. The team immediately proposes buying an online credit-recovery program. Data show most failures come from missing assignments in the first six weeks. What should the team do before buying anything?

- A.** Conclude the failures are due to students’ home lives.
- B.** Buy the credit-recovery program, since failures are the problem.
- C.** Investigate why assignments go missing early, such as transition supports or grading practices, using several data sources.
- D.** Use stadium revenue to fund tutoring.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A school team with a new data dashboard drafts the goal: “We will improve reading.” Last year 48% of third graders met the reading benchmark. The dashboard is color-coded by grade. Which revision is a SMART goal?

- A.** “All third graders will meet the benchmark by next month.”
- B.** “Third graders will read much better.”
- C.** “We will check the color-coded dashboard every week.”
- D.** “The percent of third graders meeting the reading benchmark will rise from 48% to 60% by the spring benchmark.”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A school improvement team ~~meeting in the newly renovated media center~~ reviews results from a spring survey asking families how welcome they feel at school. ~~The survey was offered in three languages.~~ What type of data is this?

Target word: classify the data **Decides:** It captures what people believe or feel.

A	Confusion set	Nothing measures what students know.
B	Correct	Surveys of opinions and feelings are perception data.
C	Camouflage	Answers the three-languages detail; the survey measures feelings, not who families are.
D	Confusion set	Process data describe what the school does, such as programs and schedules.

2. Answer A Concept 2

A fourth-grade teacher ~~who recently finished a master's degree~~ wants to decide which students need reteaching on fractions tomorrow. She has last spring's state test results, today's exit tickets, and the district's demographic report, ~~which was just updated~~. Which data source fits the decision best?

Target word: best data for tomorrow's grouping **Decides:** A next-day teaching decision needs current formative data.

A	Correct	Current formative data show who needs reteaching now.
B	Absolute language	Data must fit the question; more is not automatically better.
C	Camouflage	New, but it does not show fraction understanding.
D	Adjacent category	Summative and months old; too broad for tomorrow's groups.

3. Answer A Concept 3

A seventh-grade teacher ~~at a school with a new bell schedule~~ reports quiz scores of 72, 75, 78, 80, 81, 83 and 20. The 20 is from a student who was absent most of the unit. Which number best describes a typical score for this class?

Target word: typical score **Decides:** One extreme low score drags the mean; use the median.

A	Correct	Sorted: 20, 72, 75, 78, 80, 81, 83; the middle value is 78.
B	Wrong stem number	The outlier itself, not a typical score.
C	Confusion set	$83 - 20 = 63$ measures spread, not typical score.
D	Adjacent category	$489 / 7 \approx 69.9$ is pulled down by the outlier.

4. Answer B Concept 3 • transfer

Using the same seven scores (72, 75, 78, 80, 81, 83, 20), the teacher removes the 20. Which statistic changes more, and by about how much?

Target word: which measure moves more **Decides:** The mean is sensitive to the outlier; the median barely moves.

A	Wrong stem number	Uses the removed score, not the change in the mean.
B	Correct	$469 / 6 \approx 78.2$; median goes from 78 to 79.
C	Inverted operation	The median moves only from 78 to 79.
D	Absolute language	Both change; the mean changes much more.

5. Answer D Concept 4

School A, ~~which has a new STEM lab,~~ had 80% of students proficient last year and 80% this year. School B went from 40% to 55%. ~~Both enroll about 600 students.~~ Which school shows more evidence of student growth?

Target word: which shows more growth **Decides:** B improved 15 points; A stayed flat.

A	Camouflage	Facilities are not growth evidence.
B	Confusion set	That is status, not growth.
C	Wrong stem number	Enrollment does not decide growth.
D	Correct	Growth is change over time, not the current level.

6. Answer C Concept 4 • transfer

A board member says, “Our highest-scoring school must have the best teaching, since 92% of its students are proficient.” What does the data leave out?

Target word: what status data cannot show **Decides:** Status alone does not show how much students improved.

A	Wrong stem number	Restates the same status figure.
B	Absolute language	Status alone cannot separate teaching from starting point.
C	Correct	High status can reflect incoming achievement, not growth.
D	Adjacent category	Size does not answer whether students grew.

7. Answer D Concept 5

Students who chose to attend a voluntary after-school tutoring program, ~~held in the recently expanded library,~~ scored 12 points higher on the spring test than those who did not attend. The principal wants to credit the program. ~~The tutors were all retired teachers.~~ What is the main caution?

Target word: caution about the claim **Decides:** Voluntary attendance lets differences between groups explain the gap.

A	Wrong stem number	The size is not the issue; the comparison is.
B	Camouflage	Tutor background is not the design flaw.
C	Inverted operation	Treats correlation as causation.
D	Correct	Self-selection means correlation, not proven causation.

8. Answer B Concept 5 • transfer

A principal wants stronger evidence that a new reading intervention causes improvement. Which design helps most?

Target word: design that supports causation **Decides:** Comparable groups, with only the intervention differing.

A	Adjacent category	Perception data do not test cause.
B	Correct	Random assignment balances other factors between groups.
C	Dropped element	Leaves out any comparison group.
D	Inverted operation	Self-selection is the source of the problem.

9. Answer C Concept 6

A parent ~~who teaches college statistics~~ sees her son's report showing "60th percentile" on a nationally normed reading test ~~taken on the school's new laptops~~. She asks whether that means he got 60% of the questions right. What is the accurate explanation?

Target word: meaning of a percentile **Decides:** A percentile compares with the norm group.

A	Adjacent category	That is a criterion-referenced interpretation.
B	Camouflage	The device does not change the meaning.
C	Correct	Percentiles are norm-referenced ranks.
D	Confusion set	Confuses percentile with percent correct.

10. Answer A Concept 6 • transfer

A state test reports a student as "Meets Standard" because her score passed the fixed cut score. What kind of interpretation is this, and what does it NOT tell you?

Target word: criterion-referenced meaning **Decides:** Compared with a fixed standard, not with other students.

A	Correct	The cut score is a fixed standard, not a comparison group.
B	Confusion set	Percentiles are norm-referenced.
C	Absolute language	One cut-score result is limited evidence.
D	Inverted operation	Reverses the two interpretations.

11. Answer C Concept 7

Ninth-grade failure rates rose this year at a high school ~~with a new football stadium~~. The team immediately proposes buying an online credit-recovery program. Data show most failures come from missing assignments in the first six weeks. What should the team do before buying anything?

Target word: step before choosing a solution **Decides:** Find the root cause behind the missing work.

A	Adjacent category	Focuses on factors outside the school's control.
B	Dropped element	Skips root-cause analysis and treats the symptom.
C	Correct	Root-cause analysis targets causes the school can control.
D	Camouflage	Answers the stadium detail, not the cause.

12. Answer D Concept 8

A school team with a new data dashboard drafts the goal: “We will improve reading.” Last year 48% of third graders met the reading benchmark. ~~The dashboard is color coded by grade.~~ Which revision is a SMART goal?

Target word: SMART revision **Decides:** Needs a group, a measure, a target and a date.

A	Absolute language	Not achievable from a 48% baseline.
B	Dropped element	No measure, target or deadline.
C	Camouflage	An activity, not a goal.
D	Correct	Specific group and measure, a baseline, a target and a deadline.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Four types of school data	School data fall into demographic (who students are), perception (what people believe), student learning (what students know), and school process (what the school does, such as programs and schedules).	Calling survey opinions or attendance at a program “achievement data.”	A data source and a question about its type.
2	Match the data to the question	Choose data whose purpose fits the decision: frequent formative checks guide next-day teaching; annual summative results judge programs and trends.	Using last spring’s state test to plan tomorrow’s lesson.	A decision with a time frame and a list of data sources.
3	Mean vs. median with outliers	The mean is pulled by extreme values; the median better describes a typical score when there are outliers.	Reporting the mean as typical when one extreme score drags it.	A list of scores with one very high or very low value.
4	Status vs. growth	Status measures show how many students reached a level at one point in time; growth measures show how much students improved over time. High status does not mean high growth.	Assuming the school with the highest proficiency rate taught the most.	Two schools or classes compared on one year’s percent proficient.
5	Correlation is not causation	Two measures moving together does not show one caused the other; other factors or selection may explain the link unless a design rules them out.	Crediting a program because participants scored higher.	Voluntary participants compared with non-participants.
6	Norm-reference d vs. criterion-referenced	A norm-referenced score (such as a percentile) compares a student with other test takers; a criterion-referenced score compares performance with a fixed standard.	Reading the 60th percentile as 60% correct.	A percentile vs. a proficiency level or cut score.
7	Root cause before action	Before choosing a solution, dig beneath the symptom to causes the school can control, using several data sources.	Jumping to a program that treats the symptom, or blaming factors outside the school’s control.	A problem statement followed immediately by a proposed purchase or program.
8	SMART goals in action plans	An action plan goal should be specific, measurable, achievable, relevant and time-bound, with a baseline and a target.	Writing a goal like “improve reading” with no measure or deadline.	A goal statement missing a number, a group, or a date.

Course scope (public description): Data Literacy and Evidence-Based Practices focuses on the development of data literacy skills educators need to improve the learning and development opportunities of K-12 students.

Candidates will practice identifying educational problems and data types, generating data, analyzing data, making inferences and drawing conclusions, and creating action plans within their educational settings. Candidates will also learn best practices for data literacy, including continuous improvement planning, approaches to professional learning communities, and instructional decision-making processes. This course has no prerequisites.